TASK: 1

Create a table listing the main functions and typical ports used by switches, routers, hubs, firewalls, and access points in a home WiFi setup.

Network Device	Main Function	Typical Ports / Interfaces
Switch	Connects multiple devices within a local area network (LAN) and forwards data to the correct device.	Ethernet ports (RJ45), SFP fiber ports, console port, power port
Router	Connects different networks and routes data between a home network and the internet.	WAN port, LAN Ethernet ports, console port (on some models), power port
Hub	Connects multiple computers in a network and broadcasts incoming data to all connected ports.	Multiple Ethernet ports (RJ45), power port
Firewall	Protects the network by filtering incoming and outgoing traffic according to security rules.	Ethernet ports, WAN/LAN interfaces, management port, power port
Wireless Access Point (AP)	Allows wireless devices such as laptops and mobile phones to connect to a wired network through Wi-Fi.	Ethernet port, PoE port (if supported), power port, antenna connectors (on some models)

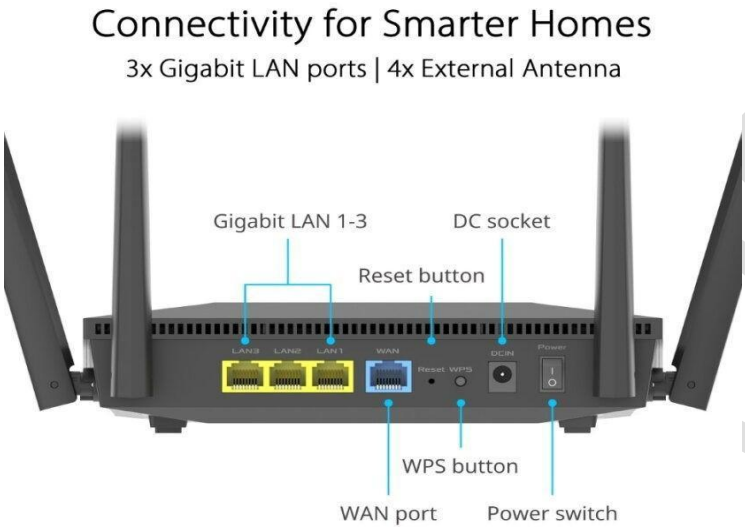
Conclusion: Each network device performs a different function. Routers connect networks, switches connect wired devices, hubs broadcast data, firewalls provide security, and access points provide wireless connectivity.

NETWORK

TASK: 2

Given images or diagrams of network devices, label each device and mark the ports (Ethernet, console, power, antenna) for at least three devices.

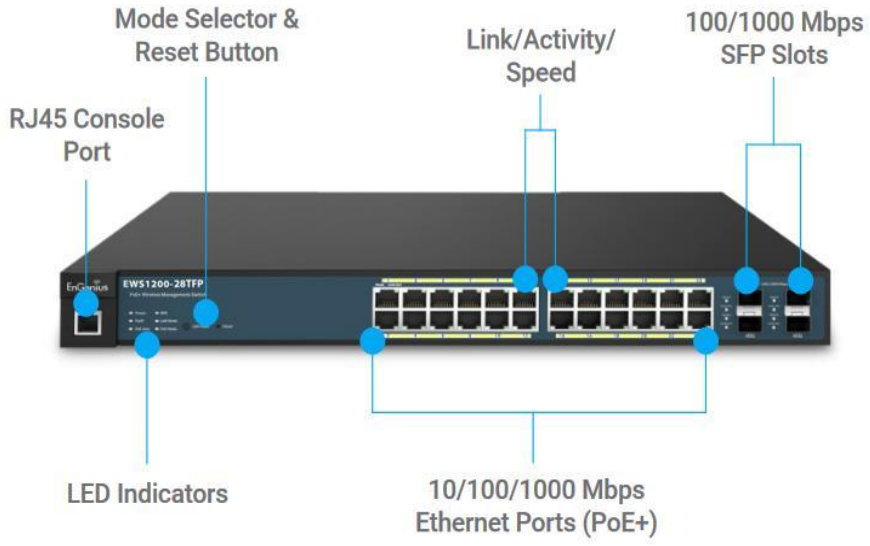
Diagram1: ROUTER



Main parts and ports:

1. **WAN port:** Connects the router to the modem or internet connection.
2. **LAN Ethernet ports:** Connect computers, switches, or other wired devices.
3. **Power port:** Supplies electrical power to the router.
4. **Antennas:** Transmit and receive wireless signals on models with external antennas.
5. **Reset button:** Restores the router to its factory settings when used according to the manufacturer's instructions.

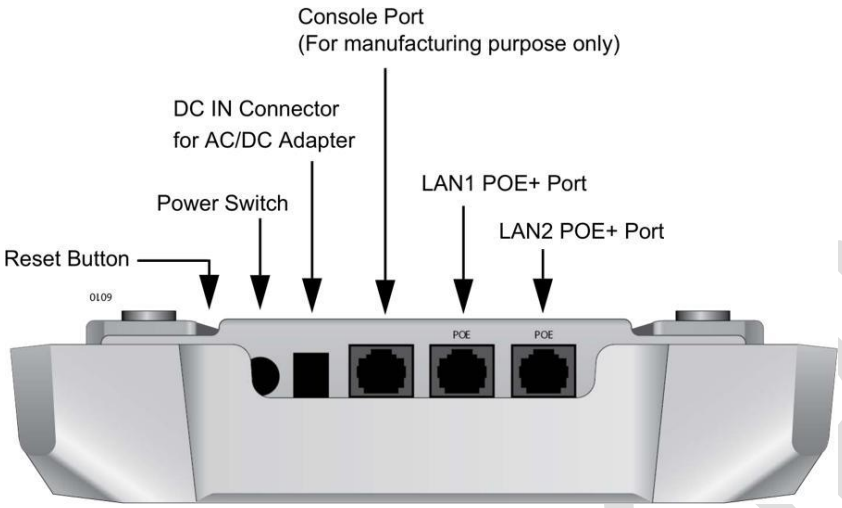
Diagram2: NETWORK SWITCH



- **Ethernet ports:** Connect computers, printers, routers, and other network devices.
- **Uplink port:** Connects the switch to another switch or router. It may be a regular Ethernet port or an SFP port.
- **Console port:** Allows administrators to configure a managed switch.

- **Power port:** Supplies electrical power to the switch.

Diagram3: WIRELESS ACCESS POINTS



Main parts and ports:

1. **Ethernet port:** Connects the access point to the wired network.
2. **PoE port:** Receives both data and electrical power through one Ethernet cable when supported.
3. **Power port:** Connects to a separate power adapter on models that support one.
4. **Antennas:** Transmit and receive Wi-Fi signals on models with external antennas.
5. **Status LED:** Shows the device's power and operating status.

Conclusion: Identifying the ports on routers, switches, and access points helps us connect and troubleshoot network devices correctly.

NETWORK

TASK: 3

Pick a popular app you use daily (like Instagram, Zomato, or Paytm) and describe in 3-4 lines how at least two different network devices (from switches, routers, firewalls, access points) play a role when you use that app.

App selected: Instagram

When I use Instagram on my mobile phone, the wireless access point connects my phone to the local network through Wi-Fi. The router forwards my data to the internet and routes incoming responses back to my device. A firewall filters network traffic according to security rules to help protect the network. Switches may also carry data between the router, access points, and other wired network devices.

NETWORK

TASK: 4

Suppose your college WiFi is slow and sometimes disconnects. Based on the features of access points, switches, and routers, suggest one possible reason for the issue and how upgrading one device could help.

Hint: Think about device capacity, coverage, and traffic handling.

One possible reason for slow college Wi-Fi is that too many students are connected to access points that cannot handle the number of devices and the amount of traffic. Weak wireless coverage or an overloaded switch or router could also contribute to the problem.

Suggested upgrade: The college could upgrade its older access points to modern, higher-capacity Wi-Fi 6 access points and install additional access points in areas with weak coverage. This would improve wireless capacity, help distribute connected devices, and provide better coverage. The college should also check that its switches and router have sufficient capacity to handle the increased network traffic.

Conclusion: Upgrading access points and checking the capacity of the router and switches can improve Wi-Fi speed, reliability, and coverage across the college.

NETWORK